

Notes: De Loecker, Eeckhout, & Unger (QJE, 2020)

The Rise of Market Power and the Macroeconomic Implications

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1 Big picture

- **Question.** How has market power evolved in the U.S. economy since the postwar period, and what macroeconomic facts can the rise in market power help explain?
- **Object measured.** The paper measures firm-level markups using the production approach: the wedge between a variable input's revenue share and that input's output elasticity.
- **Data.** The main panel is Compustat, covering publicly traded U.S. firms from 1950 to 2016; main markup figures are reported from 1955 to 2016. The paper also uses U.S. Economic Census data for manufacturing, retail, and wholesale as a robustness check.
- **Core fact.** Revenue-weighted aggregate markups are roughly flat until 1980, then rise from about 1.21 to about 1.61 by 2016.
- **Distributional fact.** The median markup is nearly unchanged. The rise is concentrated in the upper tail, especially the 90th percentile.
- **Reallocation fact.** Aggregate markups rise not only because high-markup firms raise markups, but also because sales reallocate toward high-markup firms. The paper emphasizes that reallocation accounts for roughly two-thirds of the increase.
- **Profitability fact.** Markups are not only higher because firms need to cover more overhead. Profit rates rise from about 1% of sales to about 8%, and market value and dividends scaled by sales also rise.
- **Macro message.** Higher markups imply lower factor shares through firms' first-order conditions. The paper links rising market power to the declining labor share, declining capital share, weaker labor reallocation, and lower dynamism.

2 Data and measurement

2.1 Main data: Compustat

The main dataset is firm-level Compustat. The advantage is long coverage and broad sectoral coverage. The limitation is selection: listed firms are large, older, more capital intensive, and more likely to be multinational.

- **Sales.** Revenue is used to weight firms when constructing the benchmark aggregate markup.
- **Variable input.** Cost of goods sold (COGS) is the main variable input expenditure. It bundles materials, intermediate inputs, labor cost, energy, and other production costs.
- **Capital.** The user cost of capital is constructed from interest rates, inflation, depreciation, and risk premia, and is applied to gross property, plant, and equipment.
- **Overhead.** Selling, general, and administrative expenses (SG&A) are used as the main overhead or fixed-cost measure.
- **Profitability and market valuation.** The data also contain accounting information and stock-market variables, allowing the authors to compare markups with profits, market value, and dividends.

Interpretation. Compustat is not a representative sample of all firms, but it contains many of the largest firms and is useful for studying aggregate market power because large firms carry high revenue weights. The paper's central results therefore focus on revenue-weighted moments and then check external validity using Census data.

2.2 Census robustness

The paper uses the Economic Census every five years for selected sectors.

- **Manufacturing.** Rich establishment-level information is available for output and inputs.
- **Retail and wholesale.** These data have sales and wage bills but less complete nonlabor input information.
- **Purpose.** Census evidence checks whether the Compustat rise in markups is an artifact of publicly traded firms.

Interpretation. The Census results are not identical in level because measurement differs, but they show the same broad pattern: the upper tail of the markup distribution rises, and high-markup firms become more important.

3 Markup model and empirical objects

3.1 The production approach

Firm i produces output with productivity Ω_{it} , variable input bundle V_{it} , and capital K_{it} :

$$Q_{it} = Q_{it}(\Omega_{it}, V_{it}, K_{it}).$$

Conditional on capital and fixed costs, the firm minimizes cost:

$$\mathcal{L}(V_{it}, K_{it}, \lambda_{it}) = P_{it}^V V_{it} + r_t K_{it} + F_{it} - \lambda_{it} \{Q(\cdot) - Q_{it}\}.$$

The first-order condition for the variable input implies:

$$\theta_{it}^v \equiv \frac{\partial Q(\cdot)}{\partial V_{it}} \frac{V_{it}}{Q_{it}} = \frac{1}{\lambda_{it}} \frac{P_{it}^V V_{it}}{Q_{it}}.$$

Since the Lagrange multiplier λ_{it} is marginal cost, the markup is:

$$\mu_{it} = \theta_{it}^v \frac{P_{it} Q_{it}}{P_{it}^V V_{it}}.$$

Interpretation. The method does not require estimating demand curves or imposing a model of conduct such as Bertrand or Cournot. It does require an output elasticity and a variable input revenue share. This is why the robustness section focuses so much on output elasticities, technologies, and input definitions.

3.2 Aggregate markup

The paper aggregates firm markups using weights m_{it} :

$$\mu_t = \sum_i m_{it} \mu_{it}.$$

The benchmark weight is the firm's sales share in the Compustat sample. The paper also reports input-weighted markups.

Interpretation. Revenue weighting answers: how large is the markup faced by the average dollar of sales? Input weighting answers a different question: how large is the markup faced by the average unit of inputs used? Because high-markup firms have high revenues relative to inputs, the two measures diverge when market power rises.

3.3 Profit rate

Let firm profit be sales minus variable input cost, capital cost, and overhead:

$$\Pi_{it} = S_{it} - P_t^V V_{it} - r_t K_{it} - P_t^X X_{it}.$$

Using the markup formula, the profit rate can be written as:

$$\pi_{it} = 1 - \frac{\theta_{st}}{\mu_{it}} - \frac{r_t K_{it}}{S_{it}} - \frac{P_t^X X_{it}}{S_{it}}.$$

Interpretation. This profit-rate object asks whether markups exceed what is needed to pay for variable inputs, capital, and overhead. It is crucial because high markups alone could be consistent with high fixed costs and zero economic profits.

3.4 Firm-level factor-share implications

For labor, the firm's first-order condition implies:

$$\frac{w_t L_{it}}{P_t Q_{it}} = \frac{\theta_{it}^L}{\mu_{it}}.$$

Interpretation. Holding technology fixed, a higher markup mechanically lowers the labor share. This gives the paper a clean link from micro markups to the macro decline in labor's share of income.

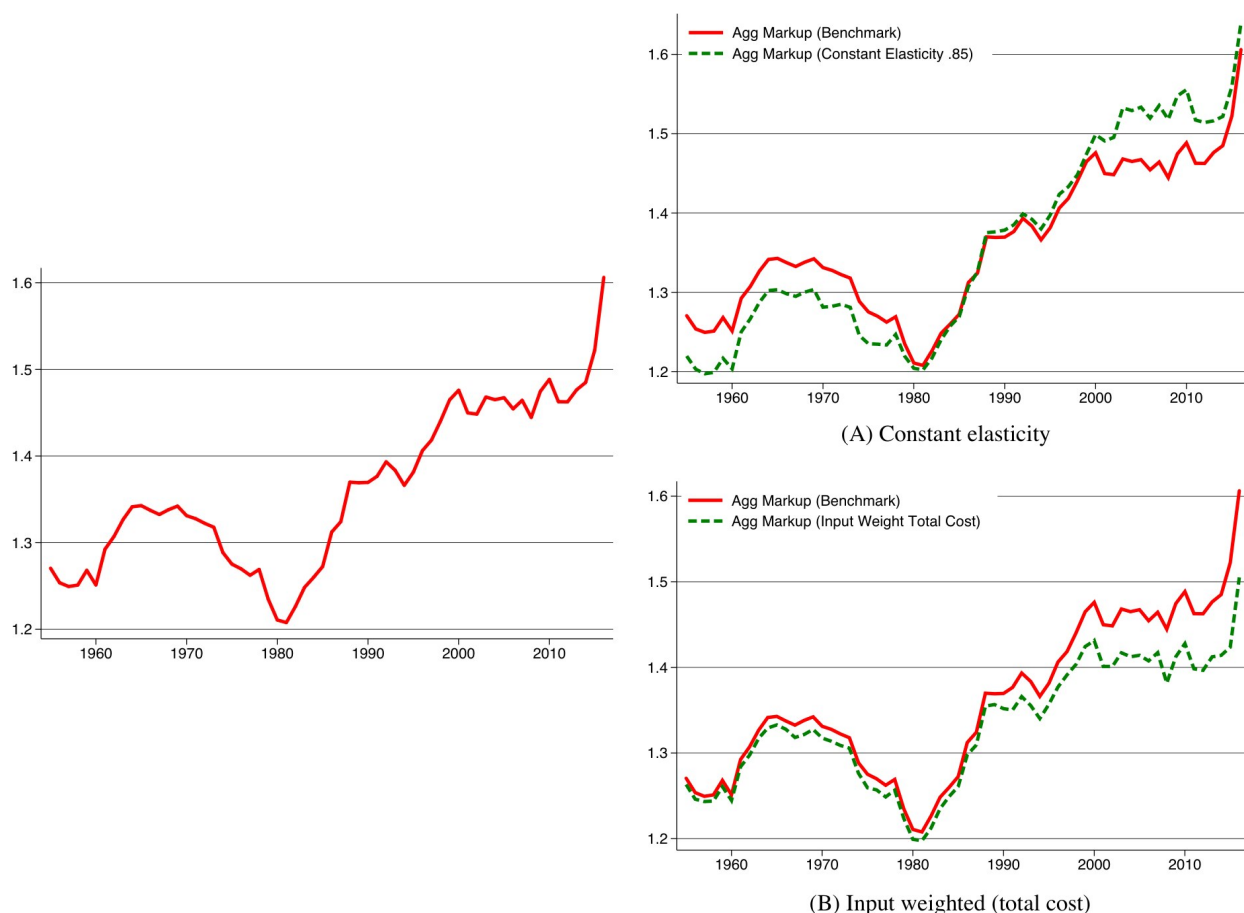
4 Identification and interpretation

- **Not a causal shock design.** The paper is primarily a measurement and macro-facts paper. It documents the rise of markups and studies consistency with several aggregate trends.
- **Key identifying assumption.** Variable inputs adjust within the year and the firm takes input prices as given. Under cost minimization, the variable input FOC identifies the markup.
- **Output elasticity risk.** If the estimated output elasticity is wrong, the markup is wrong. The paper therefore uses production-function estimates, cost-share approaches, overhead-as-factor specifications, and several weighting schemes.
- **Selection risk.** Compustat contains listed firms. Census data and sectoral reweighting are used to check whether the facts are driven by the composition of public firms.
- **Overhead interpretation.** SG&A may include selling expenses, administration, advertising, R&D, and intangible investment. The paper studies whether higher markups merely cover this overhead or generate excess profits.

Interpretation. The main empirical claim is not that every firm has more market power. It is that the distribution of market power has become much more skewed, with economic activity reallocating toward high-markup firms.

5 Figures and tables

5.1 Figures I–II: aggregate markup and alternative weights

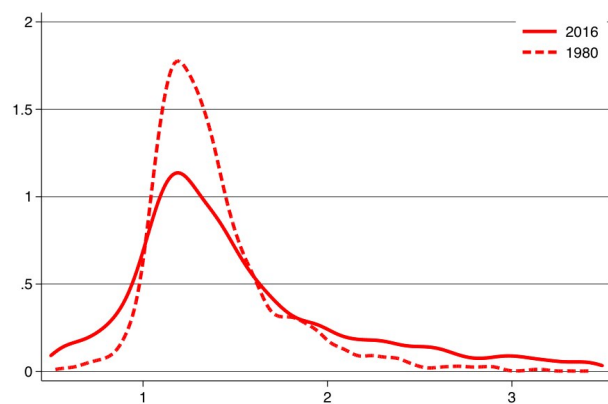


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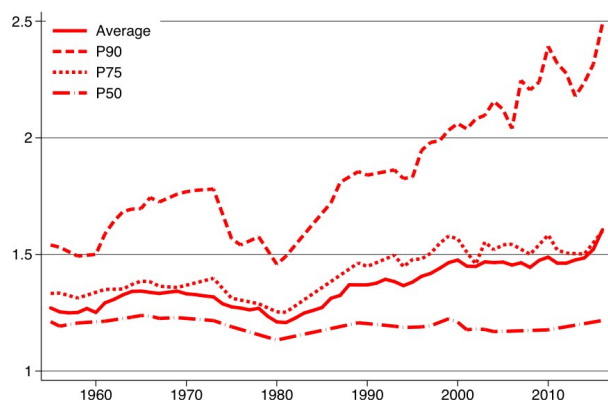
- Figure I shows the benchmark revenue-weighted markup: roughly stable before 1980, then rising from about 1.21 to about 1.61 by 2016.
- Figure II shows that the time pattern is not driven mainly by time-varying output elasticities: the constant-elasticity series has a similar rise.
- Input-weighted markups rise less than revenue-weighted markups, showing that high-markup firms command high revenue relative to inputs.

Economic message: The rise in markups is an aggregate sales-weighted phenomenon, not merely an artifact of changing production-function elasticities.

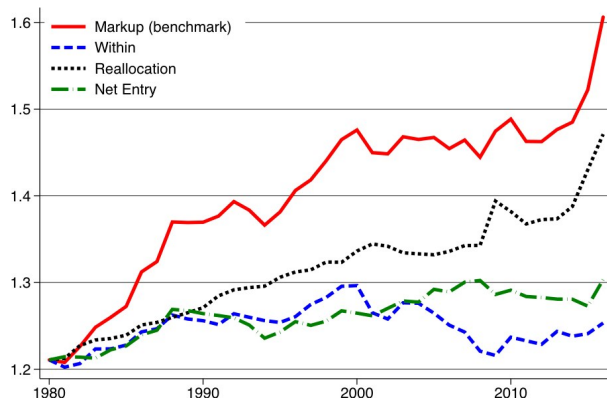
5.2 Figures III–IV: distribution and reallocation



(A) Kernel density (unweighted)



(B) Percentiles markup distribution (revenue weight)



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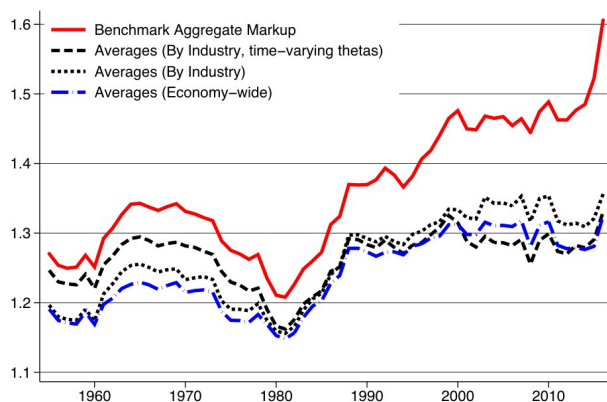
- The median markup and lower percentiles are roughly stable.
- The upper tail becomes much fatter. The 90th percentile rises sharply, from around 1.5 in 1980 to roughly 2.5 by 2016.
- The decomposition shows both within-firm/within-market increases and reallocation toward high-markup firms.

Economic message: Average markups rise because the economy becomes more dominated by a small set of high-markup firms. A representative-firm interpretation is misleading.

5.3 Table I and Figure V: aggregation matters

TABLE I
SECTORAL DECOMPOSITION OF 10-YEAR CHANGE IN MARKUP

	Markup	Δmarkup	Δwithin	$\Delta\text{between}$	Δcross
1966	1.337	0.083	0.057	-0.017	0.041
1976	1.270	-0.067	-0.055	0.002	-0.014
1986	1.312	0.042	0.035	0.010	-0.003
1996	1.406	0.094	0.098	0.004	-0.008
2006	1.455	0.049	0.046	0.007	-0.005
2016	1.610	0.154	0.133	0.014	0.007

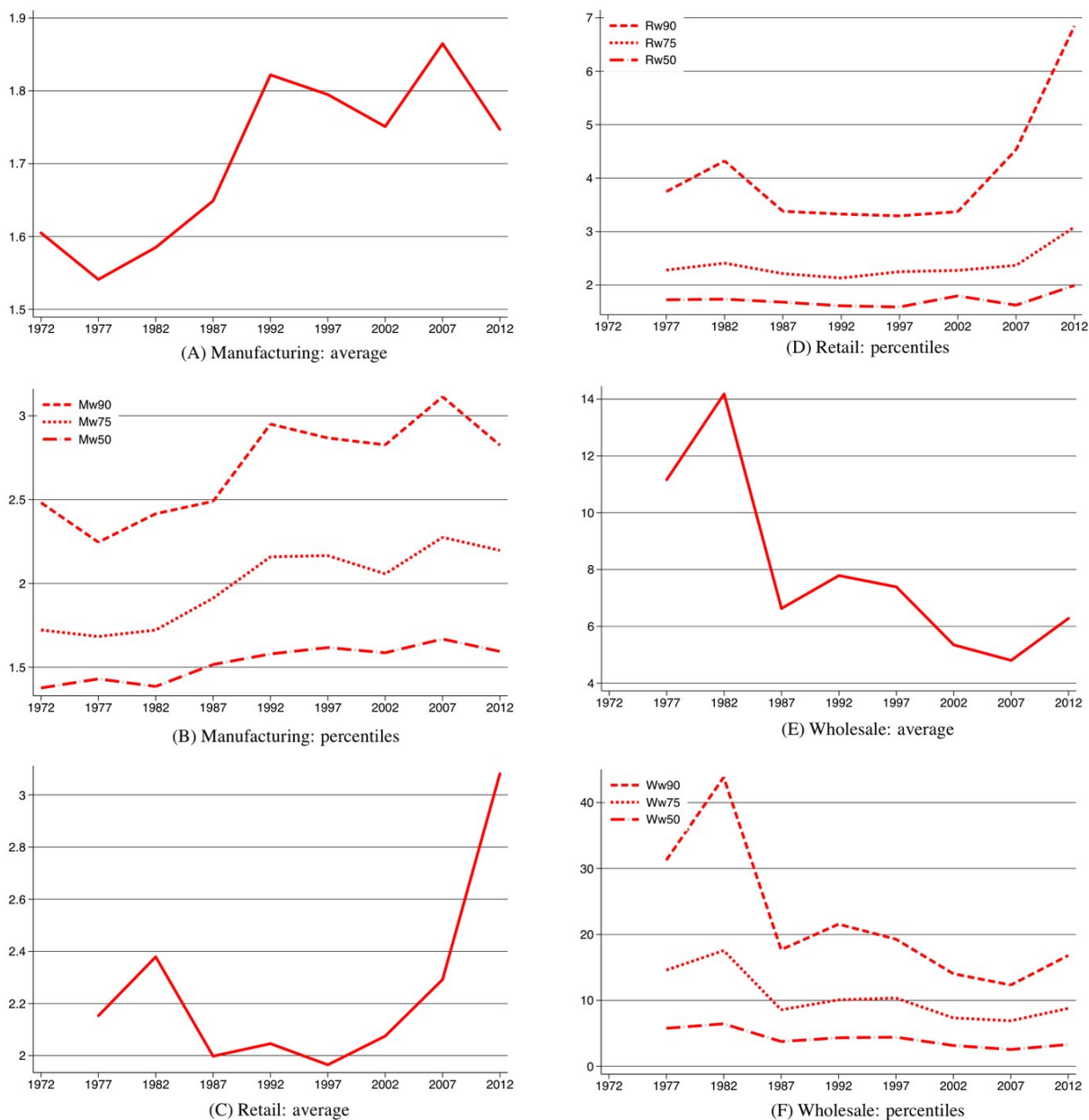


Read:

- Table I decomposes markup growth into within, reallocation, and entry/exit terms.
- Figure V shows that aggregating microdata directly differs from using economy-wide or industry-level averages.
- The within-industry reallocation component is central.

Economic message: Market power is not just an industry-composition story. The key movement is inside industries: sales shares move toward firms with higher markups.

5.4 Figure VI: Census evidence

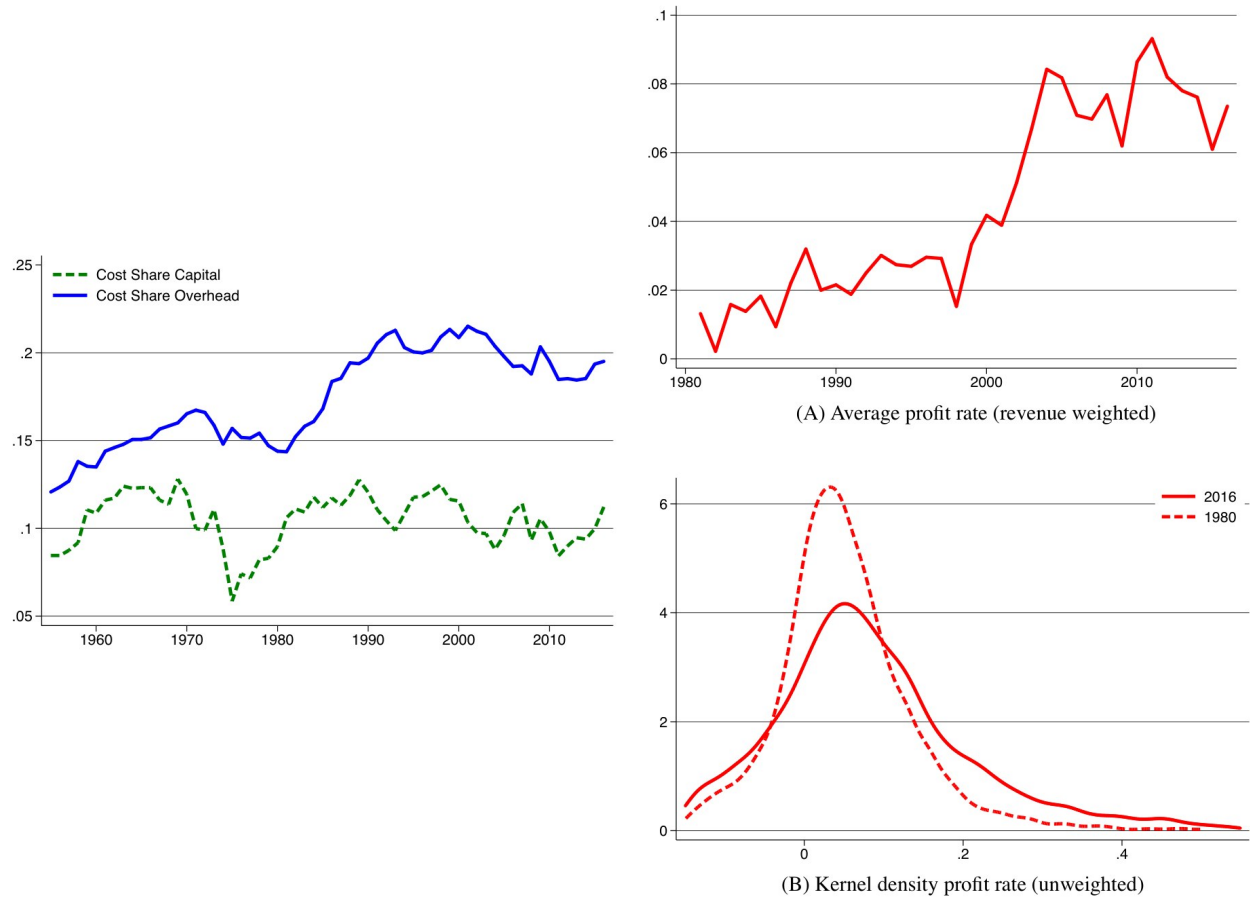


Read:

- The figure repeats the analysis using Census data for manufacturing, retail, and wholesale.
- The Census results show rising high-percentile markups, especially in the upper tail.
- The pattern is not limited to the set of publicly traded firms.

Economic message: The Compustat sample is imperfect, but the same qualitative distributional pattern appears in broader Census data.

5.5 Figures VII–VIII: overhead, capital cost, and profits



Read:

- Overhead as a share of total cost rises, roughly from 15% to 21%.
- Capital cost share fluctuates, but the key increase is not enough to absorb the markup rise.
- The average profit rate rises by about 7 percentage points between 1980 and 2016, from close to 1% to about 8%.
- The profit-rate distribution shifts in its upper tail, paralleling the markup distribution.

Economic message: Higher markups are not only compensation for higher overhead. They are associated with higher economic profits.

5.6 Figure IX and Table II: market value and dividends

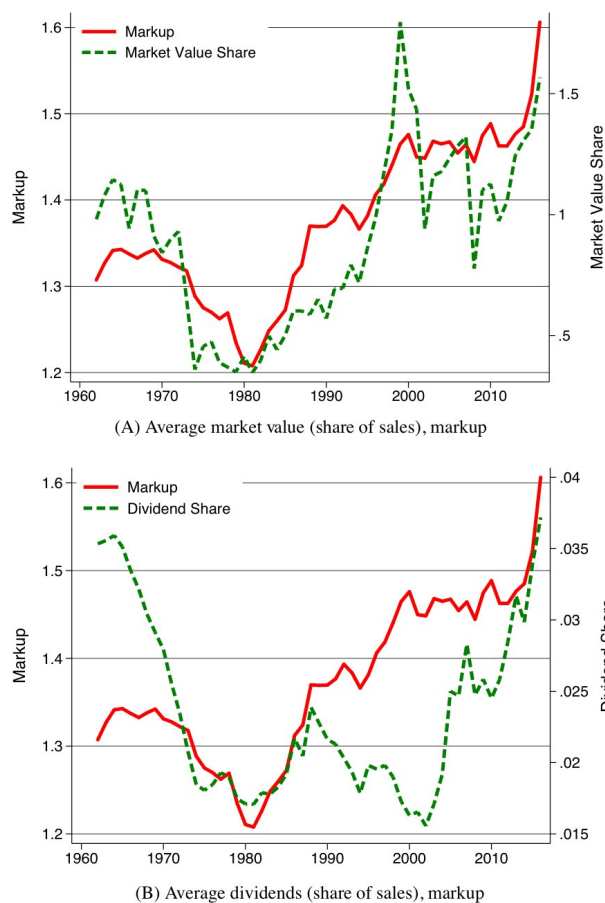


TABLE II
FIRM-LEVEL REGRESSIONS: MARKET VALUES AND DIVIDENDS ON MARKUPS

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	$\ln\left(\frac{\text{market value}}{\text{sales}}\right)$				$\ln(\text{market value})$			
$\ln(\text{markup})$	0.71 (0.03)	0.64 (0.02)	0.56 (0.02)	0.17 (0.03)	0.71 (0.02)	0.65 (0.02)	0.58 (0.02)	0.27 (0.02)
$\ln(\text{sales})$					0.81 (0.00)	0.81 (0.00)	0.83 (0.00)	0.68 (0.01)
Year fixed effects		Y	Y	Y		Y	Y	Y
Sector fixed effects			Y				Y	
Firm fixed effects				Y				Y
R^2	0.05	0.13	0.21	0.68	0.68	0.71	0.73	0.89
	$\ln\left(\frac{\text{dividends}}{\text{sales}}\right)$				$\ln(\text{dividends})$			
$\ln(\text{markup})$	1.05 (0.04)	0.97 (0.03)	0.80 (0.04)	0.26 (0.05)	1.03 (0.04)	0.93 (0.04)	0.78 (0.04)	0.26 (0.05)
$\ln(\text{sales})$					0.94 (0.01)	0.92 (0.01)	0.93 (0.01)	0.76 (0.02)
Year fixed effects		Y	Y	Y		Y	Y	Y
Sector fixed effects			Y				Y	
Firm fixed effects				Y				Y
R^2	0.06	0.11	0.17	0.70	0.66	0.68	0.70	0.89

Read:

- Market value scaled by sales rises from below 50% in 1980 to above 150% in 2016.
- Dividends as a share of sales rise from about 1.7% in 1980 to above 3.5% in 2016.
- Firm-level regressions show that markups are positively related to market value and dividends, even with fixed effects.

Economic message: Stock-market evidence supports the interpretation that rising markups represent rising market power, not just accounting noise.

5.7 Figure X and Table III: SG&A, excess markups, R&D, and advertising

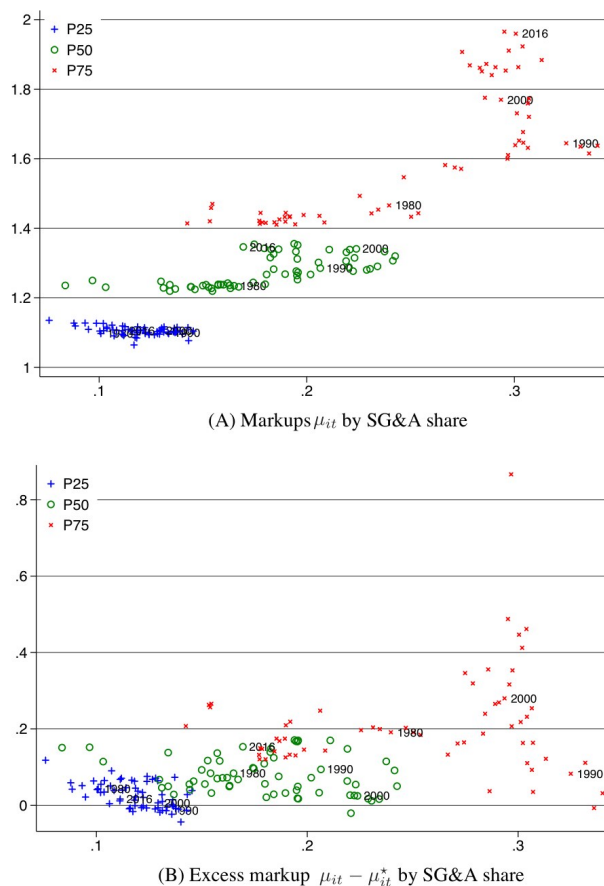


TABLE III
REGRESSIONS: EFFECT OF SG&A, R&D EXPENDITURE, AND ADVERTISING
EXPENDITURE ON MARKUPS AND PROFIT RATE; EXTENSIVE MARGIN EFFECT OF R&D
AND ADVERTISING

	Markup (log)			Profit rate (log)	
	(1)	(2)	(3)	(4)	(5)
SG&A (log)	0.56 (0.01)			0.15 (0.03)	
R&D exp. (log)		0.16 (0.01)			0.10 (0.01)
Advertising exp. (log)		0.05 (0.00)			0.03 (0.01)
R&D dummy			0.06 (0.01)		
Advertising dummy			-0.00 (0.03)		
R^2	0.61	0.07	0.43	0.04	0.05
N	26,743		247,615	26,743	

Read:

- Figure X sorts firms by SG&A share and shows that high-overhead firms also charge high markups.
- The paper defines a zero-profit markup and studies the excess markup above that benchmark.
- Table III shows positive relations between SG&A, R&D, advertising, markups, and profit rates.

Economic message: Intangibles and overhead are part of the story, but they do not eliminate the market-power interpretation: firms with more overhead also have higher excess markups and profits.

5.8 Figure XI and Table IV: why aggregate profits are not 38%

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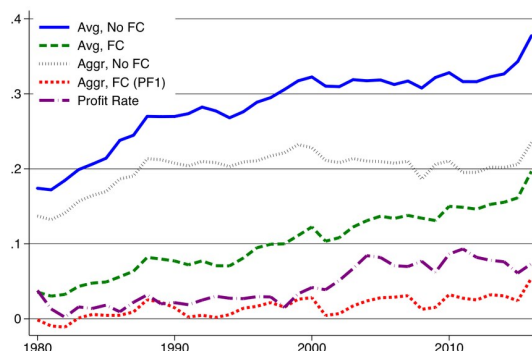


FIGURE XI

Decomposition of Equation (15) due to Overhead Costs and Aggregation

TABLE IV

DECOMPOSITION OF THE AVERAGE PROFIT RATE FROM EQUATION (15)

		Profit rate	
		1980	2016
Use average	No fixed cost	0.17	0.38
Use average	Fixed cost	0.04	0.20
Aggregation	No fixed cost	0.14	0.24
Aggregation	Fixed cost	0.00	0.05

representative firm and a constant average to marginal cost ratio

Read:

- A naive representative-firm calculation with no fixed cost implies an unrealistically high profit rate.
- Adding fixed costs reduces the implied profit rate.
- Correct aggregation across heterogeneous firms reduces the implied profit rate further.

Economic message: Markup aggregation is nonlinear. High average markups do not mechanically imply representative-firm profits because firms differ in size, cost structure, and markup.

5.9 Tables V–VI: factor-share implications

TABLE V
REGRESSIONS: LOG (LABOR SHARE) ON LOG (MARKUP)

	Labor share (log)					
	(1)	(2)	(3)	(4)	(5)	(6)
Markup (log)	−0.24 (0.03)	−0.23 (0.03)	−0.20 (0.03)	−0.24 (0.03)	−0.68 (0.02)	−0.73 (0.02)
Cost share (log)					0.91 (0.01)	0.96 (0.01)
Year FE		X	X	X	X	X
Industry FE			X		X	
Firm FE				X		X
R ²	0.02	0.08	0.21	0.88	0.93	0.99
N				24,838		

Note: FE = fixed effects. Four-digit industries. Standard errors (in parentheses) are clustered at the firm level.

TABLE VI
REGRESSIONS: LOG(CAPITAL SHARE) ON LOG(MARKUP)

	Capital share (log)					
	(1)	(2)	(3)	(4)	(5)	(6)
Markup (log)	0.03 (0.02)	0.03 (0.02)	−0.02 (0.01)	−0.14 (0.02)	−0.90 (0.00)	−0.86 (0.00)
Cost Share (log)					1.13 (0.00)	1.11 (0.00)
Year FE		X	X	X	X	X
Industry FE			X		X	
Firm FE				X		X
R ²	0.00	0.02	0.31	0.83	0.98	1.00
N				242,692		

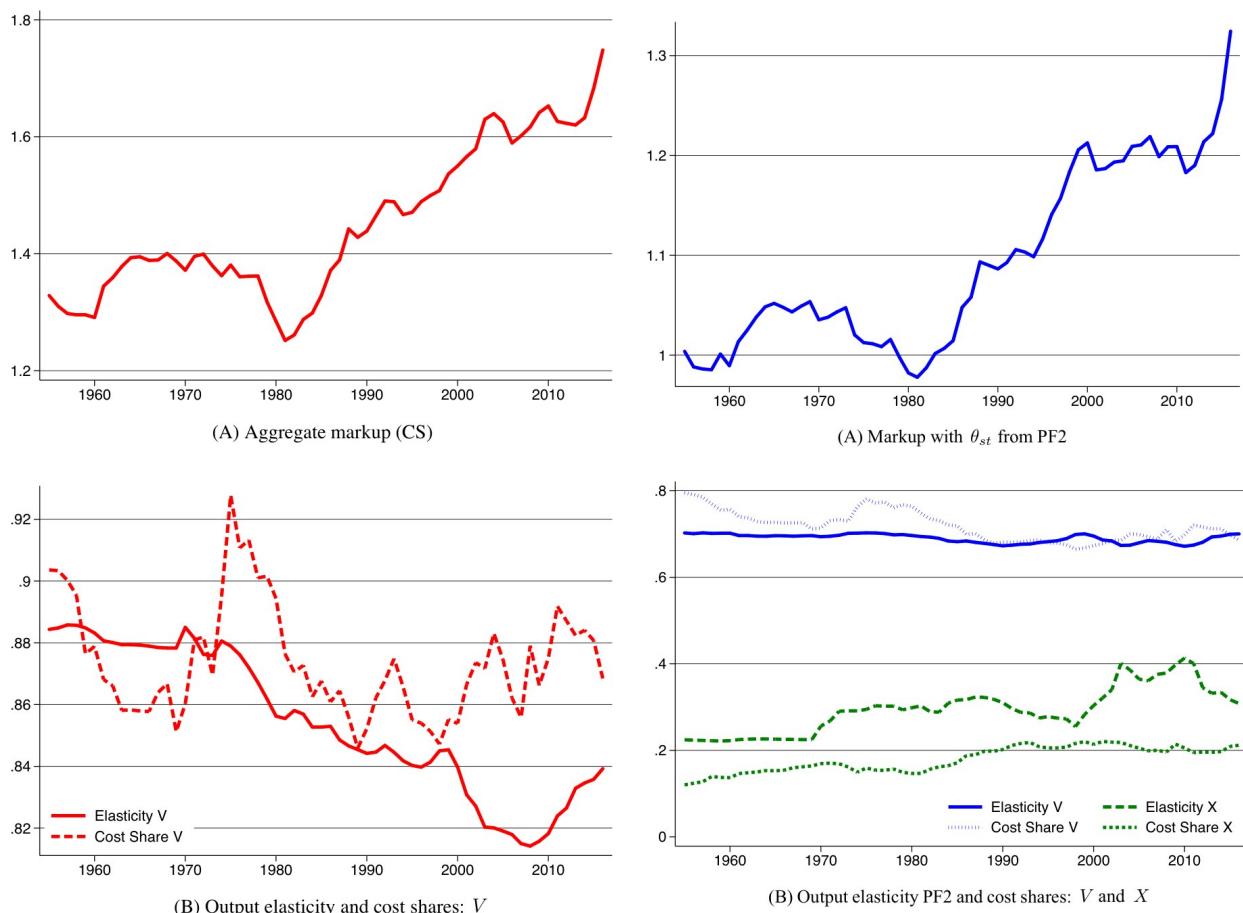
Note: FE = fixed effects. Four-digit industries. Standard errors (in parentheses) are clustered at the firm level.

Read:

- Table V regresses log labor share on log markup. The basic specifications find coefficients around -0.20 to -0.24 .
- With cost-share controls, the markup coefficient remains negative and significant.
- Table VI shows a weaker and more adjustment-cost-sensitive relation for capital shares; with firm fixed effects, the markup coefficient is negative.

Economic message: The micro evidence matches the first-order-condition logic: higher markups are associated with lower factor shares.

5.10 Figures XII–XIII: cost-share and overhead-as-factor robustness

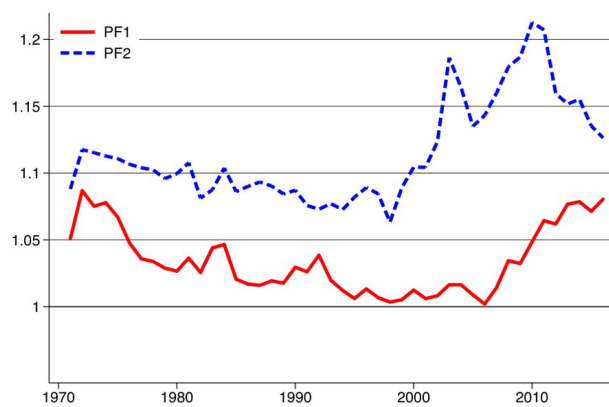


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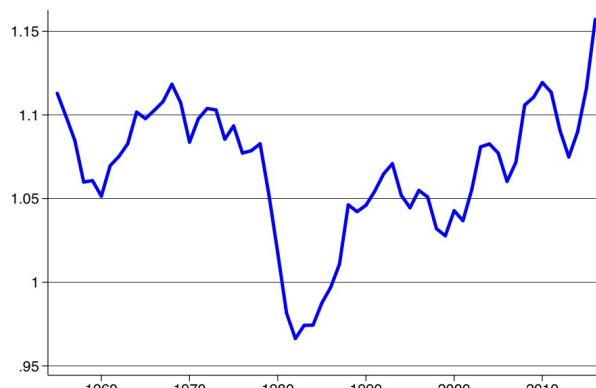
- Figure XII compares cost-share-based markups with the benchmark production-function approach.
- Figure XIII treats overhead as a factor of production rather than purely as a fixed cost.
- The broad post-1980 rise in markups remains visible under these alternatives.

Economic message: The result is not tied to a single production-function implementation.

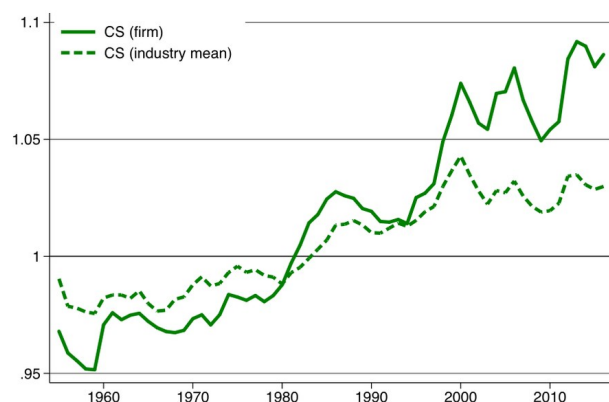
5.11 Figures XIV–XV: returns to scale and overhead technology



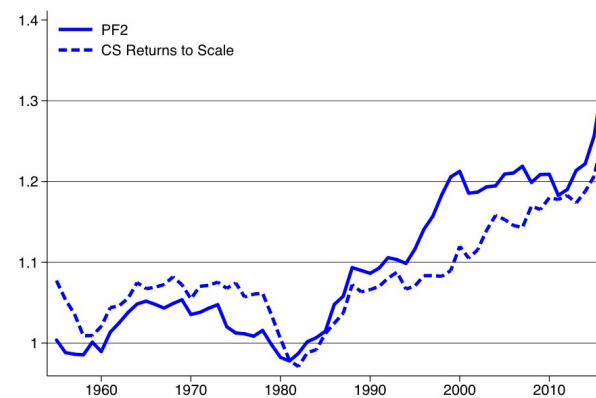
(A) Returns to scale (sum of output elasticities) of estimated PF1 and PF2; revenue weighted



(A) Markup with θ_{st} from CS2



(B) Estimated returns to scale of cost shares: firm CS and sector average CS



(B) Average markup: benchmark (PF2) and cost shares with returns to scale (Syverson)

Read:

- Figure XIV studies returns to scale, especially when overhead is included as a production factor.
- Figure XV shows average markups under the overhead-factor production function and related cost-share approaches.
- These tests address the concern that rising overhead or increasing returns mechanically create the markup result.

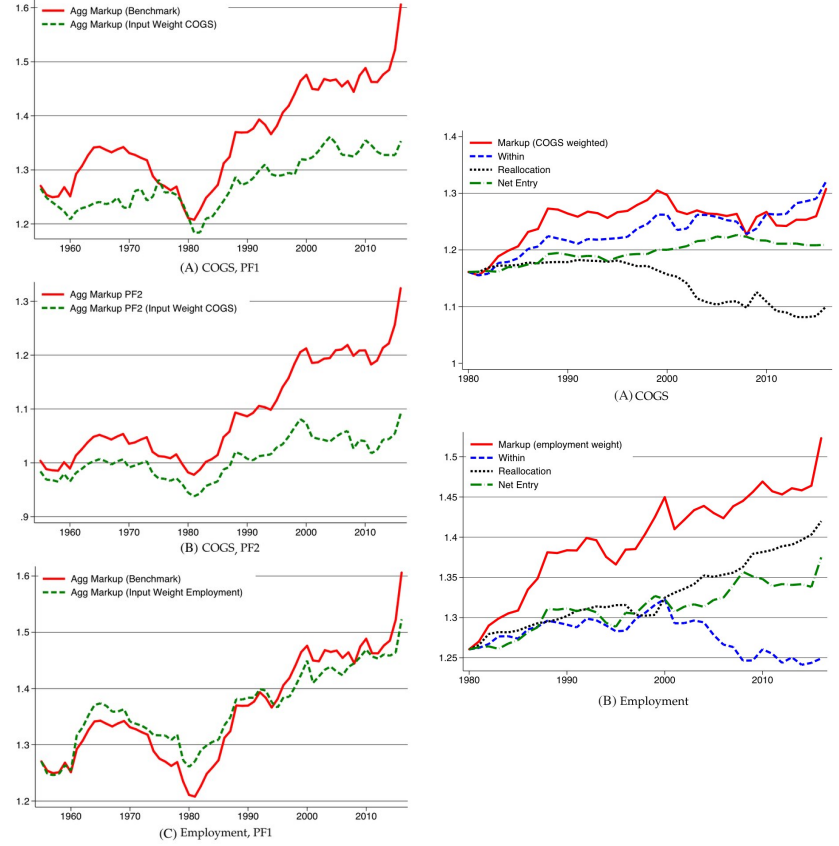
Economic message: Allowing overhead to be productive and allowing returns to scale does not overturn the main market-power facts.

5.12 Table VII and Figures XVI–XVII: weighting and joint distributions

TABLE VII
AGGREGATE MARKUPS: VARIATION BY TECHNOLOGY (θ) AND WEIGHTING (m_{it})

Output elasticity	Revenue weight	Input weight			
		1 Input		All inputs	
	$m_{it} = \frac{p_{it}}{p_c}$	$m_{it} = \frac{COGS_{it}}{COGS_c}$	$m_{it} = \frac{L_{it}}{L_c}$	$m_{it} = \frac{TC_{it}}{TC_c}$	
Economy-wide θ	$\theta \sum m_{it} \frac{p_{it}}{p_c}$	$\theta \frac{p_{it}}{COGS_c}$	$\theta \sum m_{it} \frac{L_{it}}{COGS_c}$	$\theta \sum m_{it} \frac{TC_{it}}{COGS_c}$	
Sector-specific θ_{it}	$\sum \theta_{it} m_{it} \frac{p_{it}}{p_c}$	$\sum \theta_{it} \frac{p_{it}}{COGS_c}$	$\sum \theta_{it} m_{it} \frac{L_{it}}{COGS_c}$	$\sum \theta_{it} m_{it} \frac{TC_{it}}{COGS_c}$	

Note: $TC_{it} = COGS_{it} + r_{it}K_{it} + SG_{it}$.



Read:

- Table VII clarifies how output elasticities and weights combine in aggregate markup formulas.
- Figure XVI uses input weights such as COGS and employment.
- Figure XVII decomposes input-weighted markups. Reallocation matters less for COGS-weighted markups but remains important for employment-weighted markups.

Economic message: Different weights reveal different moments of the joint distribution of markups, revenues, and inputs. The benchmark revenue-weighted rise is economically meaningful because high-markup firms also command high sales.

6 Short summary

- The paper measures markups using the production approach, not a demand-system approach.
- The benchmark U.S. revenue-weighted average markup rises from about 1.21 in 1980 to about 1.61 in 2016.
- The median markup is stable; the upper tail rises sharply.
- Sales reallocate toward high-markup firms, accounting for a large share of the aggregate increase.
- Higher markups are accompanied by higher profits, market values, and dividends.
- Overhead and intangible-type spending rise, but they do not fully explain the markup increase.
- Firm-level evidence links higher markups to lower labor shares and, with adjustment frictions, lower capital shares.
- Robustness checks with Census data, alternative output elasticities, overhead-as-factor production functions, and input weights preserve the main message.

7 Conclusion

7.1 Core conclusion

The U.S. economy has experienced a major post-1980 rise in market power. The increase is concentrated in the upper tail of the firm markup distribution and is amplified by reallocation of sales toward high-markup firms.

7.2 Economic intuition

If a firm can charge a higher price relative to marginal cost, it needs fewer inputs per dollar of sales. When high-markup firms become large, the aggregate economy shifts toward lower labor and capital shares and higher profits.

7.3 Takeaway

The paper's most important message is distributional rather than representative-firm. Most firms do not experience a dramatic markup increase. The macroeconomy changes because a small group of high-markup firms becomes more dominant, and their markups, profits, and sales shares jointly rise.